Direction-Preserving Fast Adversarial Training: Improving the Robustness-Accuracy Trade-off and Stability

Anonymous ECCV 2026 Submission

Paper ID #6070

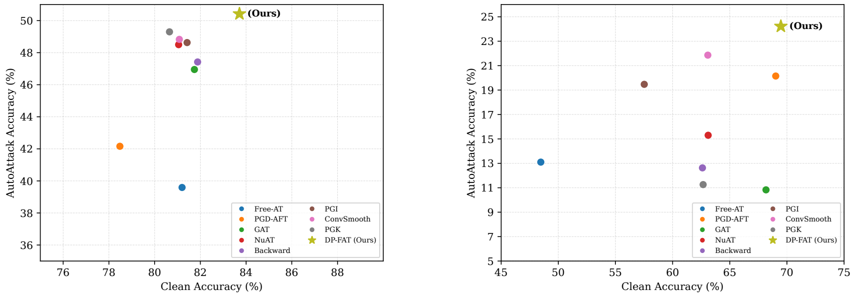
Abstract. Fast adversarial training significantly reduces computational costs but often suffers from catastrophic overfitting and a severe degradation in clean accuracy. In this paper, we propose Direction-Preserving Fast Adversarial Training (DP-FAT), which effectively harnesses the knowledge of natural models to stabilize and enhance fast AT. We identify that the high confidence of natural models often relies on non-robust features, providing misleading supervisory signals during distillation. To resolve this, we introduce Direction-Preserving Distillation, which adaptively scales teacher outputs to filter out unreliable magnitudes while selectively extracting essential class-discriminative directions. Extensive experiments show that DP-FAT achieves a superior robustness-accuracy trade-off, yielding 83.70% clean accuracy and 50.43% AutoAttack accuracy on CIFAR-10, outperforming state-of-the-art fast AT methods by significant margins. Notably, our method maintains remarkable stability under large perturbations ($\epsilon = 16/255$) where existing methods often collapse. Our approach also scales to Tiny ImageNet and ImageNet, establishing a new state-of-the-art for efficient and reliable adversarial defense.

Keywords: Adversarial Training · Fast Adversarial Training · Catastrophic Overfitting · Knowledge Distillation

1 Introduction

Adversarial training (AT) is widely recognized as the most reliable defense against adversarial attacks [1,25]. However, the standard AT framework requires generating strong adversarial examples through multi-step iterative optimization (e.g., PGD) at each training step. This iterative process incurs an immense computational burden, making it notoriously difficult to scale to large datasets and complex architectures. To alleviate this bottleneck, Fast Adversarial Training (FAT) methods have been actively investigated to achieve robustness with significantly reduced computational overhead.

Early FAT approaches, such as FGSM-RS [36], successfully accelerates the training process by utilizing single-step attacks combined with random initialization. Subsequent studies [5,19,20] have further refined these fast approximation techniques through novel loss designs to improve robustness while maintaining



(a) $\epsilon = 8/255$ **(b)** $\epsilon = 16/255$

Fig. 1: Comparison of Clean vs. Robust accuracy under different ϵ. Our DP-FAT effectively improves the performance trade-off between clean and robust accuracy and achieves State-of-the-Art (SOTA) results at both (a) $\epsilon = 8/255$ and (b) $\epsilon = 16/255$. Notably, DP-FAT maintains higher clean accuracy while ensuring stability at larger ϵ where other FAT baselines fail.

training efficiency. Despite this computational advantage, existing FAT methods fundamentally suffer from two critical limitations. First, they are highly susceptible to catastrophic overfitting —a phenomenon where the model overfits to the weak adversarial examples, causing a sudden and complete collapse of robustness. This instability becomes particularly severe under larger perturbation budgets, such as $\epsilon = 16/255$ [40]. Second, FAT methods inherently struggle with a severe degradation in clean accuracy. While the trade-off between clean and robust performance is a well-known dilemma in standard AT [8, 30], effectively bridging this accuracy gap within the constrained computational budget of the fast training regime remains a formidable challenge.

In this paper, we propose a novel FAT framework that fully leverages a pre-trained natural model to resolve these issues. Our method is built upon two key strategies: First, natural model initialization accelerates convergence, enabling the use of stronger multi-step attacks without sacrificing training efficiency. Crucially, this combination ensures remarkable training stability, maintaining a consistent optimization trajectory even under extreme perturbations (e.g., large ϵ) where existing fast AT methods catastrophically overfit. Second, to close the remaining clean accuracy gap, we utilize the knowledge of the pre-trained natural model through knowledge distillation, rather than relying on it solely for model initialization. However, naively distilling this knowledge severely degrades the student's robustness. This occurs because natural models inherently rely on non-robust shortcuts [11, 16], producing excessively overconfident outputs that manifest as abnormally large feature norms. To safely extract the essential semantic knowledge without transferring these brittle features, we introduce Direction-Preserving Distillation (DPD). DPD adaptively scales down the overconfident feature magnitudes of the natural model for each sample while preserving its discriminative directions. This approach effectively minimizes the influence of non-robust shortcuts while inheriting the teacher's superior classification performance.

By integrating this distillation strategy into the accelerated optimization process, we propose our unified framework: **Direction-Preserving Fast Adversarial Training (DP-FAT)**. Notably, DP-FAT establishes a self-distillation pipeline where the pre-trained natural model serves a dual role: it provides the initial weights to facilitate rapid convergence and acts as the teacher to provide continuous semantic guidance. As shown in Figure 1, our proposed framework, DP-FAT, achieves a superior balance of clean and robust accuracy across multiple benchmarks. Furthermore, our method is approximately 30% faster in training time than conventional FAT baselines on average. Our main contributions are summarized as follows:

- We propose a two-stage framework DP-FAT that first leverages natural model initialization to significantly accelerate training convergence, and subsequently introduces DPD to selectively transfer discriminative directions while filtering out unreliable magnitudes.
- We demonstrate the superior scalability of our method through extensive experiments on CIFAR-10, CIFAR-100, Tiny-ImageNet, and ImageNet, achieving State-of-the-Art (SOTA) performance across all benchmarks.
- Our framework ensures stability at large perturbations ($\epsilon = 16/255$) where existing FAT methods often collapse, while cutting training time by 30–40%.

2 Related Work

The most representative AT method, Projected Gradient Descent (PGD) AT [25], employs the PGD attack to generate adversarial examples. To further increase robustness, several works have proposed additional regularization losses [2,21,31–34,37–39,41–43]. For instance, consistency loss, which encourages similar outputs for clean and adversarial inputs, has been widely utilized [8,20,34,38]. Furthermore, stochastic weight averaging (SWA) has shown to be effective in smoothing the loss landscape for better robustness [6,17,20]. Despite these advancements, AT inherently suffers from a fundamental trade-off between clean and robust accuracy. To address this inevitable degradation of clean performance, several works leverage the knowledge of highly accurate natural models [8,30]. Specifically, LBGAT [8] constrains the robust model's logits to match those of a natural model, while ARREST [30] applies knowledge distillation within a fine-tuning framework.

However, due to the immense computational cost of standard AT, numerous FAT methods have been proposed to enhance training efficiency. Free-AT [27] updates model parameters and perturbations in a single backward pass, while Guided Adversarial Training (GAT) [28] uses a relaxation term to guide the initial perturbation. Other efficiency improvements include NuAT [29] using Nuclear-Norm regularization, Backward [5] proposing inner maximization initialization, and memory-based methods like PGI and PGK [19,20] that recycle historical perturbations. Building on this, ConvSmooth [40] further stabilizes these fast approaches at large perturbation boundaries. PGD-adversarial fine-tuning (PGD-AFT) [18] fine-tunes a pre-trained natural model using a cyclic learning rate schedule, focusing primarily on improving the training efficiency of AT.

Diverging from these methods that target either accuracy or efficiency in isolation, our approach comprehensively leverages the pre-trained natural model to achieve both: significantly accelerating training convergence while strictly preserving high clean accuracy through adaptive distillation.

3 Analysis

3.1 Notation

Let $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$ be a dataset of N samples. We define a classifier $f(\boldsymbol{x}) = W^T \Phi(\boldsymbol{x})$, where $\Phi(\cdot)$ is a feature extractor and W is a linear classifier. Adversarial training (AT) is formulated as the following minimax optimization:

$$\min_{\theta} \mathbb{E}_{(\boldsymbol{x},y)\sim\mathcal{D}} \left[\max_{\boldsymbol{x}_{adv} \in \mathcal{B}(\boldsymbol{x},\epsilon)} \mathcal{L}(f(\boldsymbol{x}_{adv}), y) \right], \quad (1)$$

where $\mathcal{B}(\boldsymbol{x}, \epsilon) = \{\boldsymbol{x}' : \|\boldsymbol{x}' - \boldsymbol{x}\|_\infty \leq \epsilon\}$ is the ϵ-perturbation ball and $\mathcal{L}$ is the loss function (e.g., cross-entropy).

3.2 Motivation

Existing fast AT methods inherently struggle with a harsh trade-off between **robust overfitting** and **suboptimal clean accuracy**. To resolve these critical limitations, we build our approach in two key steps. First, we revisit the structural stability and efficiency provided by natural initialization. Second, we dissect the confidence reliability of natural models to establish the empirical basis for our Direction-Preserving Distillation.

(I) Stability and Efficiency through Natural Initialization. Fine-tuning from a pre-trained natural model, as explored in PGD-AFT [18], provides a well-conditioned landscape rich in pre-learned features. Because the model is already partially optimized, it requires significantly fewer training epochs to converge. This drastically reduced training duration frees up the computational budget, enabling the use of reliable multi-step attacks (e.g., PGD) while still retaining the overall speed of fast AT. Despite this dual advantage, such fine-tuning approaches are often overlooked as their robust accuracy typically lags behind state-of-the-art fast AT methods. However, we observe that the true value of natural initialization lies in its remarkable training stability, especially under extreme conditions. As illustrated in Figure 2, while existing fast AT methods such as Backward [5] and PGK [20] suffer from

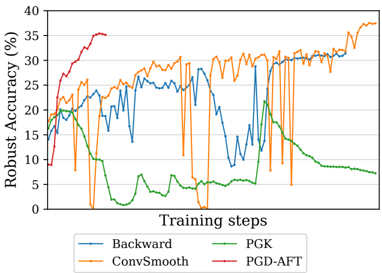


Fig. 2: Training chart against a large perturbation $\epsilon = 16/255$.

Table 1: Comparison of accuracy (%) and logit entropy between natural models and robustly trained models on CIFAR-100.

Metric	Natural Models			Robust Models		
	Natural 1	Natural 2	Natural 3	LTD [4]	BDM-AT [35]	IKL-AT [9]
Clean Acc	96.03	96.13	95.27	64.07	72.58	73.80
AutoAttack Acc	0.00	0.00	0.00	30.57	38.83	39.18
Entropy	0.048	0.049	0.081	2.96	2.74	2.26

catastrophic overfitting at $\epsilon = 16/255$, the natural-initialized model maintains a consistent training trajectory. This stability stems directly from the affordable integration of multi-step adversaries, which inherently prevent the failure modes of single-step generation. Motivated by this, we adopt the pre-trained natural model as the initialization point for our framework to secure a reliable optimization trajectory.

(II) Reliability of Confidence in Natural Models.

While natural initialization addresses training instability, the challenge of closing the clean accuracy gap remains. A straightforward approach is to leverage the superior representation of a natural model through knowledge distillation [8,30]. However, a naive application of knowledge distillation often compromises adversarial robustness, as the student may inherit the teacher's unreliable features.

To understand whether the confidence of a natural teacher is a reliable signal for a robust student, we ground our analysis in the framework of [16], which distinguishes between *robust features* (consistently predictive under attack) and *non-robust features* (highly predictive but sensitive to small perturbations). Prior work [16] shows that standard natural models tend to rely heavily on non-robust features, which can yield highly confident predictions despite being fragile under adversarial perturbations. In Table 1, we can see that natural models exhibit significantly lower entropy and substantially higher confidence compared to robust models. Based on this, we formulate the following conjecture regarding the underlying mechanics of model confidence.

Conjecture 1. The over-reliance on non-robust features in natural models is inherently linked to feature magnitude. Specifically, natural models exploit brittle shortcuts to maximize prediction confidence, a process that manifests as disproportionately large feature norms and leads to over-confident, near-zero entropy distributions.

To validate this conjecture and confirm that high-magnitude features are indeed inherently brittle, we investigate the latent space of the natural teacher model f_t under adversarial attacks. Specifically, we measure *feature deviation*, defined as the L_2 distance between clean and adversarial representations ($\|\Phi_t(\boldsymbol{x}) - \Phi_t(\boldsymbol{x}_{adv})\|_2$). As Figure 3 shows, this deviation strictly increases as the teacher's feature norm grows (Q1 to Q5). This confirms that abnormally large norm features heavily rely on non-robust shortcuts, making them highly vulnerable.

To further understand how these high-magnitude, non-robust features affect the distillation process, we consider a knowledge distillation framework where the student f_s aims to align its adversarial representation with the teacher's clean one. Specifically, the distillation objective is formulated as minimizing the loss $\mathcal{L}_{KD}(f_s(\boldsymbol{x}_{adv}), f_t(\boldsymbol{x}))$. To quantify how strongly each sample contributes to this alignment process, we compute the gradient of the distillation loss with respect to the student feature. Since the gradient magnitude reflects how strongly a sample drives the representation update during distillation, analyzing feature gradient reveals which samples dominate the distillation dynamics. As evidenced in Figure 3, we observe a strong positive correlation between the feature magnitude and the resulting KD gradient norm. This indicates that the most vulnerable samples—those with the largest feature deviations—dominate the optimization dynamics during distillation. Consequently, matching absolute logit values allows these excessive gradients to bias the optimization process, forcing the student to prioritize non-robust features and brittle shortcuts over stable semantic knowledge.

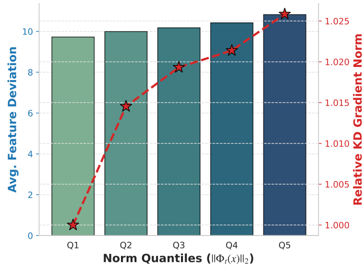


Fig. 3: Empirical validation of Conjecture 1. Higher teacher feature norms exhibit strictly monotonic increases in both feature deviation (bars) and relative KD gradient norm (dashed line) under PGD attacks, indicating severe structural instability and subsequent optimization disruption.

Nevertheless, natural models still contain valuable semantic knowledge. The challenge is therefore to transfer this knowledge without inheriting the unreliable confidence signals induced by non-robust features. This motivates our approach: instead of matching absolute logit values, we focus on the *relative class preferences* encoded in the logit direction, naturally suppressing magnitude-driven instabilities.

4 Proposed Method: Direction-Preserving Fast AT

In this section, we present our fast AT framework designed to exploit the strengths of natural models while neutralizing their inherent unreliability.

4.1 Natural Model Initialization

As established in Section 3.2, initializing the adversarial training process from a pre-trained natural solution provides a well-conditioned landscape that significantly accelerates convergence. We initialize the student model parameters θ with the weights of a natural model θ_{nat}. This initialization not only allows for a much shorter training epochs but also ensures that the student starts near a high-accuracy manifold, which is then refined for robustness.

4.2 Direction-Preserving Distillation (DPD)

While natural initialization provides a good starting point, maintaining the clean accuracy requires continuous guidance throughout the fast adversarial training process. To this end, we adopt a **self-teaching** strategy: we clone the naturally initialized model to serve as a static teacher f_t, which possesses rich semantic knowledge but lacks adversarial robustness.

However, as established in Conjecture 1, a natural model's absolute confidence is often an unreliable signal driven by non-robust shortcuts. Therefore, to prevent robustness collapse, it is imperative to explicitly penalize the influence of these unreliable, magnitude-driven signals.

Sample-adaptive Distillation. To filter out these structural instabilities while distilling the teacher's knowledge, we propose DPD. For a clean input $\boldsymbol{x}$ and its corresponding adversarial example $\boldsymbol{x}_{adv}$, we define a **sample-adaptive temperature** $\kappa(\boldsymbol{x})$ that rescales the teacher's directional logit based on its instantaneous feature norm on the clean data:

$$\kappa(\boldsymbol{x}) = 1 + \tau \|\Phi_t(\boldsymbol{x})\|_2, \tag{2}$$

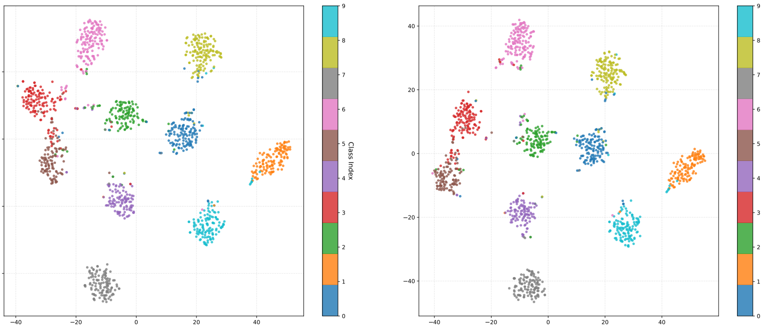
where τ is a scaling hyperparameter controlling the penalty strength. The DPD loss aligns the student's normalized prediction on the adversarial example with the teacher's adaptive guidance on the clean example:

$$\mathcal{L}_{DPD} = \mathrm{KL}\left(\sigma(W_s^T \hat{\boldsymbol{z}}_s) \,\|\, \sigma\left(\frac{W_t^T \hat{\boldsymbol{z}}_t}{\kappa(\boldsymbol{x})}\right)\right) \tag{3}$$

where $\hat{\boldsymbol{z}}_s = \Phi_s(\boldsymbol{x}_{adv})/\|\Phi_s(\boldsymbol{x}_{adv})\|_2$ and $\hat{\boldsymbol{z}}_t = \Phi_t(\boldsymbol{x})/\|\Phi_t(\boldsymbol{x})\|_2$ denote the L_2-normalized directional features for the student and the teacher, respectively, and $\sigma(\cdot)$ is the softmax function, $\mathrm{KL}(\cdot\|\cdot)$ denotes the Kullback-Leibler divergence.

This formulation reveals the inherent adaptivity of DPD. By embedding the feature norm $\|\Phi_t(\boldsymbol{x})\|_2$ directly into the temperature term $\kappa(\boldsymbol{x})$, the model dynamically adjusts the softening effect for each input. The base temperature of 1 ensures that normal samples with moderate norms retain their informative, sharp confidence. Conversely, samples with abnormally large magnitudes are assigned a higher temperature, which proportionally dampens their disproportionate influence and prevents the student from inheriting these non-robust shortcuts.

Analysis of Directional Semantics. One might wonder whether penalizing feature magnitude leads to the loss of essential semantic information. To address this concern, we visualize the teacher's latent space. As shown in Figure 4, the cluster structure of the features transformed by our sample-adaptive DPD formulation Figure 4b remains remarkably consistent with the raw features Figure 4a. To quantitatively validate this visual observation, we evaluate the predictive performance on the CIFAR-10 test set using these DPD-adjusted representations. Remarkably, the clean accuracy experiences a negligible drop of merely 0.11% (from 94.51% to 94.40%). This compelling empirical evidence confirms



(a) Raw Features **(b)** Directional Features

Fig. 4: t-SNE visualization of teacher features. A comparison between raw features (z_t) and directional features ($\frac{\hat{\boldsymbol{z}}_t}{\kappa(\boldsymbol{x})}$) reveals that the categorical cluster structure is remarkably well-preserved even after L_2 normalization. This justifies our DPD approach: we can safely extract essential semantic information ("what to predict") from the directional features, allowing us to utilize the non-robust feature magnitude strictly as a penalty signal.

that highly discriminative categorical information ("what to predict") is inherently preserved despite the magnitude penalty. Therefore, it thoroughly justifies our approach: we can safely distill the stable semantic direction ($\hat{\boldsymbol{z}}_t$) while utilizing the non-robust feature magnitude strictly as a penalty signal.

4.3 Overall Training Objective

To finalize our DP-FAT framework, we supplement the proposed $\mathcal{L}_{DPD}$ with a self-consistency regularization to enforce predictive invariance. Specifically, we employ a logit consistency loss $\mathcal{L}_C$ that minimizes the discrepancy between the student's outputs on clean ($\boldsymbol{x}$) and adversarial ($\boldsymbol{x}_{adv}$) examples:

$$\mathcal{L}_C = \mathrm{KL}\left(\sigma(f_s(\boldsymbol{x}_{adv})) \,\|\, \sigma(f_s(\boldsymbol{x}))\right). \tag{4}$$

During the distillation process, the student is strictly optimized to extract stable semantic directions from the adversarial inputs, guided by the teacher's representations of the clean inputs. Thus, our overall training objective $\mathcal{L}_{total}$ is elegantly formulated as:

$$\mathcal{L}_{total} = \mathcal{L}_{DPD} + \lambda \cdot \mathcal{L}_C, \tag{5}$$

where λ is a hyperparameter controlling the regularization strength. Note that $\mathcal{L}_{DPD}$ is computed asymmetrically using the student's adversarial features ($\hat{\boldsymbol{z}}_s$) and the teacher's clean features ($\hat{\boldsymbol{z}}_t$).

Finally, following recent practices in fast adversarial training [20], we apply Stochastic Weight Averaging (SWA) to the student model's parameters throughout the fine-tuning process to further enhance robust generalization.

Table 2: Clean and robust accuracy (%) against ℓ_∞ perturbations of $\epsilon = 8/255$ on CIFAR-10 using ResNet-18. Training time is recorded in hours.

Method	Clean	FGSM	PGD-10	PGD-20	PGD-50	C&W	AA	Time
Free-AT [27]	81.20	50.07	44.34	43.20	42.84	41.88	39.59	2.54
PGD-AFT [18]	78.48	52.11	47.55	46.53	46.38	44.80	42.16	**0.40**
GAT [28]	81.74	58.68	53.50	52.52	52.21	49.28	46.95	1.27
NuAT [29]	81.05	58.02	53.32	52.37	52.00	50.71	48.50	1.16
Backward [5]	<u>81.88</u>	58.38	52.54	51.29	51.02	49.70	47.42	1.17
PGI [19]	81.42	58.80	55.15	54.39	54.24	50.66	48.63	1.21
ConvSmooth [40]	81.08	58.84	55.51	54.81	54.60	50.49	48.83	1.20
PGK [20]	80.65	<u>59.00</u>	<u>55.57</u>	<u>55.34</u>	<u>55.28</u>	<u>50.84</u>	<u>49.30</u>	1.33
DP-FAT	**83.70**	**60.36**	**55.98**	**55.80**	**55.30**	**52.42**	**50.43**	<u>0.88</u>

Table 3: Clean and robust accuracy (%) against ℓ_∞ perturbations of $\epsilon = 8/255$ on CIFAR-100 using ResNet-18. Training time is recorded in hours.

Method	Clean	FGSM	PGD-10	PGD-20	PGD-50	C&W	AA	Time
Free-AT [27]	51.84	25.69	22.45	21.87	21.75	19.99	17.64	2.53
PGD-AFT [18]	54.71	25.46	22.39	21.87	21.75	19.87	17.75	**0.40**
GAT [28]	<u>59.81</u>	31.76	28.60	28.09	27.93	24.75	22.35	1.27
NuAT [29]	59.31	29.47	26.48	22.88	20.45	21.34	13.55	1.17
Backward [5]	58.67	32.96	30.02	29.48	29.39	26.34	24.43	1.17
PGI [19]	59.01	33.55	30.81	30.30	30.29	26.63	24.84	1.20
ConvSmooth [40]	58.15	<u>34.10</u>	<u>31.71</u>	<u>31.25</u>	<u>31.13</u>	27.12	25.09	1.23
PGK [20]	58.77	33.53	30.74	30.18	30.09	<u>27.38</u>	<u>25.42</u>	1.33
DP-FAT	**60.40**	**35.66**	**33.65**	**33.31**	**33.26**	**28.38**	**26.63**	<u>0.88</u>

5 Experiments

5.1 Experimental Settings

Training. We conducted experiments on CIFAR-10/100 [22], Tiny ImageNet [23], and ImageNet [10]. We utilized a three-step PGD [25] with a step size set to half of the ϵ boundary, where $\epsilon = 8/255$. We used ResNet-18 [13] on CIFAR-10/100, PreActResNet-18 [14] on Tiny ImageNet, and ResNet-50 [13] on ImageNet.

Evaluation. We evaluated clean and robust accuracy under FGSM [12], PGD (10, 20, and 50 steps) [25], C&W [3], and AutoAttack (AA) [7]. All training times were measured on a single NVIDIA GeForce RTX 3090 GPU. For a fair comparison, the reported training time for our method explicitly includes the pre-training phase of the natural model, representing the total end-to-end execution time from scratch.

Baseline defense. We selected several state-of-the-art methods: Free-AT [27], PGD-AFT [18], GAT [28], NuAT [29], Backward [5], PGI [19], ConvSmooth [40], and PGK [20]. We adhered closely to their reported training hyperparameters. In cases where detailed hyperparameter settings were not provided, we reported the best results obtained through grid search.

Table 4: Clean and robust accuracy (%) against ℓ_∞ perturbations of $\epsilon = 8/255$ on Tiny ImageNet using PreActResNet-18. Training time is recorded in hours.

Method	Clean	FGSM	PGD-10	PGD-20	PGD-50	C&W	AA	Time
Free-AT [27]	50.50	19.44	15.81	15.09	14.77	14.18	12.08	16.32
PGD-AFT [18]	49.62	20.98	18.32	17.83	17.68	15.64	13.58	**2.59**
GAT [28]	51.57	22.42	19.03	18.45	18.41	15.78	13.36	8.15
NuAT [29]	44.51	22.12	19.98	19.65	19.59	16.26	14.97	6.58
Backward [5]	47.59	24.36	22.18	21.87	21.86	17.87	16.55	7.36
PGI [19]	47.70	22.26	19.59	19.21	19.13	14.94	13.22	6.23
ConvSmooth [40]	47.64	22.30	19.71	19.24	19.17	15.01	14.03	7.55
PGK [20]	43.26	24.16	23.05	22.89	22.87	18.30	17.01	6.65
DP-FAT	**52.16**	**29.10**	**27.12**	**26.75**	**26.73**	**21.82**	**19.78**	4.07

5.2 Main Results

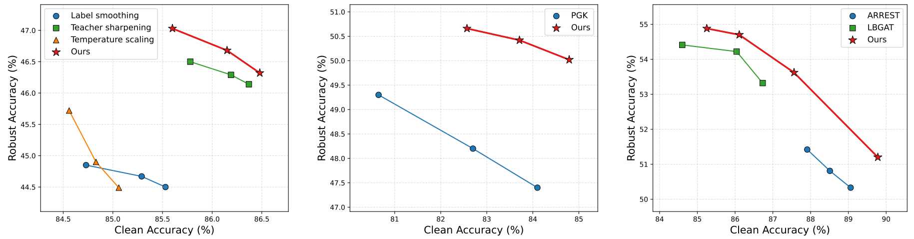
The experimental results on CIFAR-10, CIFAR-100, Tiny ImageNet, and ImageNet are summarized in Table 2 to Table 5. Overall, our proposed DP-FAT consistently achieves the best performance in both clean and robust accuracy across all benchmarks while maintaining superior time efficiency. Furthermore, DP-FAT exhibits high stability with an overall standard deviation of approximately 0.1 across multiple runs (see supplementary material for detailed variance analysis).

Performance on CIFAR and Tiny ImageNet. On CIFAR-10 and CIFAR-100, DP-FAT is approximately 33% faster in training time than PGK [20], the leading state-of-the-art baseline. Furthermore, we observe that PGK exhibits a notable degradation in clean accuracy on Tiny ImageNet compared to earlier baselines such as Free-AT [27] or GAT [28]. DP-FAT successfully overcomes this limitation, achieving a 8.90%p improvement in clean accuracy over PGK alongside superior defense capabilities against diverse adversarial attacks.

Scalability to ImageNet. The advantages of DP-FAT are even more pronounced on large-scale datasets. As shown in Table 5, our method delivers a substantial leap in performance on ImageNet. Specifically, DP-FAT outpaces PGK by nearly **3× in training speed** while simultaneously enhancing clean accuracy by 4.89%p and robust accuracy (AA) by 1.2%p. These results demonstrate that DP-FAT provides a highly scalable and practical solution, effectively mitigating the robustness-efficiency trade-off as dataset complexity increases.

Table 5: ImageNet results ($\epsilon = 4/255$, ResNet-50).

Methods	Clean	AA	Time
Free-AT [27]	60.42	24.47	131.82
Backward [5]	52.53	26.95	133.76
PGK [20]	60.42	28.04	102.31
DP-FAT	**65.31**	**29.24**	**36.59**



(a) (b) (c)

Fig. 5: Analysis of clean–robust Pareto frontiers on CIFAR-10, where robust accuracy is evaluated against AA. (a) DPD vs. conventional softening strategies. (b) Trade-off comparison with PGK [20]. (c) Comparison with natural-model-guided standard AT methods by varying trade-off parameters.

5.3 Ablation Studies

Effectiveness of Sample-Adaptive Softening. A natural question is whether DPD is simply another variant of conventional confidence softening strategies. To rigorously examine this, we compare DPD against three representative baselines by sweeping their respective softening parameters τ to obtain Pareto frontiers. Specifically, we consider *standard label smoothing* ($\mathcal{L}_{LS}$), which redistributes a fraction τ of the ground-truth probability mass [26]. We also evaluate *global temperature scaling* ($\mathcal{L}_{TS}$) [15], which softens both student and teacher distributions using a constant temperature τ, formulated as $\mathrm{KL}(\sigma(f_s(\boldsymbol{x}_{adv})/\tau) \parallel \sigma(f_t(\boldsymbol{x})/\tau))$. Furthermore, we include *teacher sharpening* ($\mathcal{L}_{Sharp}$), where τ is applied *exclusively* to the teacher's logits to modulate its confidence, formulated as $\mathrm{KL}(\sigma(f_s(\boldsymbol{x}_{adv})) \parallel \sigma(f_t(\boldsymbol{x})/\tau))$. In contrast to these fixed or uniform strategies, our DPD employs a *sample-adaptive* τ that specifically penalizes high-norm features. For a fair comparison, we isolate the softening effect by employing only the KL divergence term, excluding consistency loss or weight averaging.

As shown in Fig. 5a, DPD consistently forms a superior Pareto frontier in the clean–robust trade-off space. Across all τ values, our method achieves higher robust accuracy at comparable clean accuracy levels. This consistent Pareto dominance indicates that the gains cannot be explained by uniform softening or global temperature manipulation. Instead, the improvement stems from DPD's representation-level constraint, which preserves semantic direction while mitigating magnitude-driven overconfidence.

Superiority in the Clean–Robust Pareto Frontier. We further compare the Pareto frontiers of DPD and PGK [20], the two most competitive methods in the main results. By systematically varying the weight of the consistency loss in both methods, which governs the trade-off between clean and robust accuracy, we trace the performance limits of both methods. As illustrated in Fig. 5b, DPD consistently maintains a superior frontier over PGK.

Comparison with Natural-Model-Guided Methods. We evaluate the scalability of DPD by comparing it against standard adversarial training baselines, *LBGAT* [8] and *ARREST* [30], which also utilize natural model guidance. Under

Table 6: Performance on CIFAR-10 under natural initialization.

Methods	Clean	PGD	C&W	AA
Free-AT	80.84	44.12	43.48	41.16
PGD-AFT	78.48	46.38	44.80	42.16
GAT	80.59	52.74	48.57	46.63
NuAT	83.37	40.06	42.01	23.43
Backward	80.58	51.06	49.45	47.65
PGI	80.36	52.21	48.71	47.04
ConvSmooth	79.96	53.05	48.85	47.18
PGK	82.87	52.39	49.03	46.91
DP-FAT	**83.70**	**55.30**	**52.42**	**50.43**

Table 7: Performance on CIFAR-10 under large perturbation ($\epsilon = 16/255$).

Methods	Clean	PGD	C&W	AA
Free-AT	48.47	18.45	17.09	13.10
PGD-AFT	69.02	27.78	23.71	20.15
GAT	68.17	15.70	15.01	10.83
NuAT	63.11	20.69	19.43	15.30
Backward	62.61	24.04	17.19	12.63
PGI	57.52	23.43	23.43	19.47
ConvSmooth	63.08	32.40	26.32	21.86
PGK	62.67	16.75	13.85	11.26
DP-FAT	**69.47**	**34.21**	**30.83**	**24.22**

a matched training budget and SAT regime, DPD consistently outperforms both methods (Fig. 5c). These results suggest that our representation-level constraint provides more effective guidance than the conventional distillation strategies of existing standard adversarial training-based methods, regardless of the training protocol. A more detailed comparison is provided in the supplementary material.

Effect of Natural Model Initialization. To ensure our gains are not merely from initialization, we evaluate baselines using the same pre-trained natural model (Table 6). Natural initialization yields marginal or no improvements for most methods and severely degrades *NuAT*. This confirms that DP-FAT's superiority stems from DPD's continuous representation-level guidance, rather than just the starting weights.

Robustness under Larger Epsilon Bounds. Following the protocol of *ConvSmooth* [40], which is specifically designed for large-epsilon regimes, we evaluate our method under more challenging robustness constraints by increasing the perturbation boundary (Table 7). Our method achieves the best performance in both clean and robust accuracy. Notably, DP-FAT surpasses the state-of-the-art by a considerable margin in PGD accuracy while maintaining higher clean accuracy.

Generalization to Distribution Shifts. While fast adversarial training methods are often criticized for their potential instability or overfitting to specific attack patterns, a robust model should generalize to unseen distribution shifts. To demonstrate the stability and reliability of our approach, we evaluate DP-FAT on *OODRobustBench* [24], which simulates diverse real-world scenarios including domain and threat model shifts. As shown in Table 8, DP-FAT consistently outperforms all baselines across both in-distribution (ID) and out-of-distribution (OOD) benchmarks. Specifically, it

Table 8: Performance under distribution shifts on OODRobustBench.

Methods	ID	OOD_d	ID	OOD_d	OOD_t
Free	81.20	69.59	39.59	28.10	23.88
PGD-AFT	78.48	68.83	42.16	29.87	25.30
GAT	81.74	71.73	46.95	33.82	28.96
NuAT	81.05	70.14	48.50	34.74	28.37
Backward	81.88	72.34	47.42	35.47	29.31
PGI	81.42	71.93	48.63	35.11	30.12
ConvSmooth	81.08	71.44	48.83	35.33	30.62
PGK	80.65	70.57	49.30	35.82	32.40
DP-FAT	**83.70**	**73.12**	**50.37**	**35.83**	**33.62**

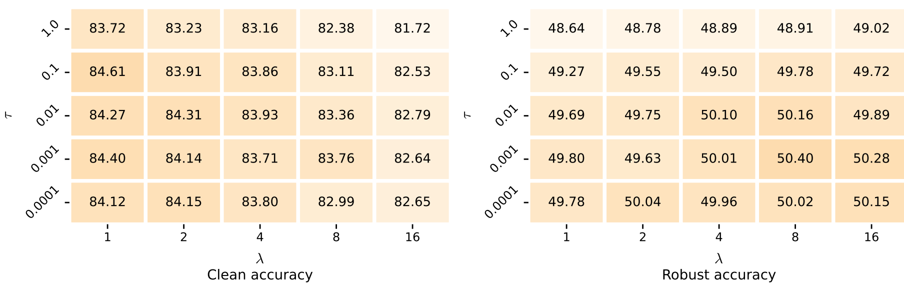


Fig. 6: Clean and robust accuracy on CIFAR-10 under various combinations of τ and λ, where robust accuracy is evaluated against AA.

achieves the best results under both domain shifts (OOD_d) and threat model shifts (OOD_t), confirming that our sample-adaptive softening does not merely memorize the training distribution. Instead, by mitigating magnitude-driven overconfidence and preserving semantic direction, our method ensures a more stable and generalizable robust optimization. This performance on OODRobust-Bench serves as strong evidence that DP-FAT provides superior reliability in distribution shifts. Detailed descriptions of the OODRobustBench components are available in the supplementary material.

Hyper-parameter Sensitivity. In Figure 6, we analyze the impact of the consistency weight λ and the scaling factor τ on the model's performance using the CIFAR-10 dataset. As anticipated, λ strictly dictates the clean-robust trade-off: increasing λ generally improves adversarial robustness at the expense of clean accuracy. This offers practitioners flexible control over the desired defensive objective, with $\lambda \approx 8$ emerging as an effective sweet spot that well-balances both metrics. On the other hand, DP-FAT demonstrates remarkable stability with respect to τ. While the framework is largely insensitive to substantial variations in τ, it achieves its peak performance and optimal balance when τ is in the range of $[0.001, 0.01]$. This overall low sensitivity to hyperparameter choices highlights the practical reliability and ease of tuning of our proposed method.

Effect of Each Component on DP-FAT. In Table 9, we investigate the individual and combined effects of the proposed DPD, consistency loss, and weight averaging. For clarity, configurations without DPD (e.g., the first row) directly distill the natural teacher's unmodified logits, formulated as $\text{KL}(\sigma(f_s(\boldsymbol{x}_{adv})) \| \sigma(f_t(\boldsymbol{x})))$. As widely observed in existing adversarial training literature, the integration of consistency loss exhibits a clear trade-off (improving robustness while sacrificing a certain degree of clean accuracy). Weight averaging demonstrates a similar trade-off but serves as a more substantial booster for overall adversarial robustness.

Most importantly, the results highlight the orthogonal and synergistic benefits of DPD. Notably, applying DPD alone to the base standard KD setting (the first vs. fifth row) significantly enhances robust accuracy without degrading clean performance (86.27% vs. 86.26%). Furthermore, DPD consistently yields

Table 9: Ablation study of DP-FAT components on CIFAR-10. The presence of each module is denoted by ✓.

DPD	Consistency Loss	Weight Averaging	Clean	PGD-50	C&W	AA
			86.27	46.84	47.87	44.38
	✓		84.58	48.30	48.41	45.13
		✓	83.18	52.84	51.61	48.69
	✓	✓	82.14	53.66	51.68	48.88
✓			86.26	49.24	49.79	46.34
✓	✓		85.13	50.92	50.67	47.51
✓		✓	84.37	53.95	52.36	49.77
✓	✓	✓	83.70	**55.30**	**52.42**	**50.43**

robust gains across all configurations, confirming that our representation-level constraint smoothly complements established robust optimization techniques. Ultimately, the integration of all three components synergistically achieves the best robustness (50.43% AA), demonstrating a highly effective and balanced framework for fast adversarial training.

6 Conclusion

We have proposed DP-FAT, a novel and highly efficient framework for fast adversarial training. Motivated by the observation that natural models often rely on non-robust features that inflate their predictive confidence, we have introduced DPD. By dynamically decoupling the reliable semantic direction from the overconfident feature magnitude, DPD acts as a self-regulating mechanism that provides stable and robust supervision without inheriting the teacher's vulnerabilities. Extensive experiments across various benchmarks, including ImageNet, demonstrate that DP-FAT achieves state-of-the-art clean and robust accuracy while significantly reducing training time compared to existing fast AT methods. Extending this representation-level guidance to standard, non-fast adversarial training regimes remains a promising direction for future work.

References

1. Athalye, A., Carlini, N., Wagner, D.: Obfuscated gradients give a false sense of security: Circumventing defenses to adversarial examples. In: International conference on machine learning. pp. 274–283. PMLR (2018)
2. Bai, Y., Zeng, Y., Jiang, Y., Xia, S.T., Ma, X., Wang, Y.: Improving adversarial robustness via channel-wise activation suppressing. arXiv preprint arXiv:2103.08307 (2021)
3. Carlini, N., Wagner, D.: Towards evaluating the robustness of neural networks. In: 2017 ieee symposium on security and privacy (sp). pp. 39–57. Ieee (2017)

4. Chen, E., Lee, C.: LTD: low temperature distillation for robust adversarial training. CoRR **abs/2111.02331** (2021), `https://arxiv.org/abs/2111.02331`
5. Chen, J., Cheng, Y., Gan, Z., Gu, Q., Liu, J.: Efficient robust training via backward smoothing. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 36, pp. 6222–6230 (2022)
6. Chen, T., Zhang, Z., Liu, S., Chang, S., Wang, Z.: Robust overfitting may be mitigated by properly learned smoothening. In: International Conference on Learning Representations (2020)
7. Croce, F., Hein, M.: Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks. In: International conference on machine learning. pp. 2206–2216. PMLR (2020)
8. Cui, J., Liu, S., Wang, L., Jia, J.: Learnable boundary guided adversarial training. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 15721–15730 (2021)
9. Cui, J., Tian, Z., Zhong, Z., Qi, X., Yu, B., Zhang, H.: Decoupled kullback-leibler divergence loss. arXiv preprint arXiv:2305.13948 (2023)
10. Deng, J., Dong, W., Socher, R., Li, L.J., Li, K., Fei-Fei, L.: Imagenet: A large-scale hierarchical image database. In: 2009 IEEE conference on computer vision and pattern recognition. pp. 248–255. Ieee (2009)
11. Geirhos, R., Jacobsen, J.H., Michaelis, C., Zemel, R., Brendel, W., Bethge, M., Wichmann, F.A.: Shortcut learning in deep neural networks. Nature Machine Intelligence **2**(11), 665–673 (2020)
12. Goodfellow, I.J., Shlens, J., Szegedy, C.: Explaining and harnessing adversarial examples. arXiv preprint arXiv:1412.6572 (2014)
13. He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 770–778 (2016)
14. He, K., Zhang, X., Ren, S., Sun, J.: Identity mappings in deep residual networks. In: Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part IV 14. pp. 630–645. Springer (2016)
15. Hinton, G., Vinyals, O., Dean, J.: Distilling the knowledge in a neural network. arXiv preprint arXiv:1503.02531 (2015)
16. Ilyas, A., Santurkar, S., Tsipras, D., Engstrom, L., Tran, B., Madry, A.: Adversarial examples are not bugs, they are features. Advances in neural information processing systems **32** (2019)
17. Izmailov, P., Podoprikhin, D., Garipov, T., Vetrov, D., Wilson, A.G.: Averaging weights leads to wider optima and better generalization. arXiv preprint arXiv:1803.05407 (2018)
18. Jeddi, A., Shafiee, M.J., Wong, A.: A simple fine-tuning is all you need: Towards robust deep learning via adversarial fine-tuning. arXiv preprint arXiv:2012.13628 (2020)
19. Jia, X., Zhang, Y., Wei, X., Wu, B., Ma, K., Wang, J., Cao, X.: Prior-guided adversarial initialization for fast adversarial training. In: European Conference on Computer Vision. pp. 567–584. Springer (2022)
20. Jia, X., Zhang, Y., Wei, X., Wu, B., Ma, K., Wang, J., Cao, X.: Improving fast adversarial training with prior-guided knowledge. IEEE Transactions on Pattern Analysis and Machine Intelligence (2024)
21. Jin, G., Yi, X., Huang, W., Schewe, S., Huang, X.: Enhancing adversarial training with second-order statistics of weights. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 15273–15283 (2022)

22. Krizhevsky, A., Hinton, G., et al.: Learning multiple layers of features from tiny images (2009)
23. Le, Y., Yang, X.: Tiny imagenet visual recognition challenge. CS 231N **7**(7), 3 (2015)
24. Li, L., Wang, Y., Sitawarin, C., Spratling, M.: Oodrobustbench: benchmarking and analyzing adversarial robustness under distribution shift. arXiv preprint arXiv:2310.12793 (2023)
25. Madry, A., Makelov, A., Schmidt, L., Tsipras, D., Vladu, A.: Towards deep learning models resistant to adversarial attacks. arXiv preprint arXiv:1706.06083 (2017)
26. Müller, R., Kornblith, S., Hinton, G.E.: When does label smoothing help? Advances in neural information processing systems **32** (2019)
27. Shafahi, A., Najibi, M., Ghiasi, M.A., Xu, Z., Dickerson, J., Studer, C., Davis, L.S., Taylor, G., Goldstein, T.: Adversarial training for free! Advances in Neural Information Processing Systems **32** (2019)
28. Sriramanan, G., Addepalli, S., Baburaj, A., et al.: Guided adversarial attack for evaluating and enhancing adversarial defenses. Advances in Neural Information Processing Systems **33**, 20297–20308 (2020)
29. Sriramanan, G., Addepalli, S., Baburaj, A., et al.: Towards efficient and effective adversarial training. Advances in Neural Information Processing Systems **34**, 11821–11833 (2021)
30. Suzuki, S., Yamaguchi, S., Takeda, S., Kanai, S., Makishima, N., Ando, A., Masumura, R.: Adversarial finetuning with latent representation constraint to mitigate accuracy-robustness tradeoff. In: 2023 IEEE/CVF International Conference on Computer Vision (ICCV). pp. 4367–4378. IEEE (2023)
31. Tack, J., Yu, S., Jeong, J., Kim, M., Hwang, S.J., Shin, J.: Consistency regularization for adversarial robustness. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 36, pp. 8414–8422 (2022)
32. Tian, Q., Kuang, K., Jiang, K., Wu, F., Wang, Y.: Analysis and applications of class-wise robustness in adversarial training. In: Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining. pp. 1561–1570 (2021)
33. Vivek, B., Babu, R.V.: Single-step adversarial training with dropout scheduling. In: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 947–956. IEEE (2020)
34. Wang, Y., Zou, D., Yi, J., Bailey, J., Ma, X., Gu, Q.: Improving adversarial robustness requires revisiting misclassified examples. In: International Conference on Learning Representations (2019)
35. Wang, Z., Pang, T., Du, C., Lin, M., Liu, W., Yan, S.: Better diffusion models further improve adversarial training. In: International Conference on Machine Learning (ICML) (2023)
36. Wong, E., Rice, L., Kolter, J.Z.: Fast is better than free: Revisiting adversarial training. arXiv preprint arXiv:2001.03994 (2020)
37. Xiao, C., Zhong, P., Zheng, C.: Enhancing adversarial defense by k-winners-take-all. arXiv preprint arXiv:1905.10510 (2019)
38. Zhang, H., Yu, Y., Jiao, J., Xing, E., El Ghaoui, L., Jordan, M.: Theoretically principled trade-off between robustness and accuracy. In: International conference on machine learning. pp. 7472–7482. PMLR (2019)
39. Zhang, J., Zhu, J., Niu, G., Han, B., Sugiyama, M., Kankanhalli, M.: Geometry-aware instance-reweighted adversarial training. arXiv preprint arXiv:2010.01736 (2020)

40. Zhao, M., Zhang, L., Kong, Y., Yin, B.: Fast adversarial training with smooth convergence. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 4720–4729 (2023)
41. Zheng, H., Zhang, Z., Gu, J., Lee, H., Prakash, A.: Efficient adversarial training with transferable adversarial examples. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 1181–1190 (2020)
42. Zhou, D., Wang, N., Gao, X., Han, B., Wang, X., Zhan, Y., Liu, T.: Improving adversarial robustness via mutual information estimation. In: International Conference on Machine Learning. pp. 27338–27352. PMLR (2022)
43. Zhou, D., Wang, N., Han, B., Liu, T.: Modeling adversarial noise for adversarial training. In: International Conference on Machine Learning. pp. 27353–27366. PMLR (2022)